

Q3-B: Find the Sylow Subgroups of D_{12}

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Preliminaries

Definition 1 (Sylow p -Subgroup). *Let G be a finite group with $|G| = p^n m$, where p is prime and $\gcd(p, m) = 1$. A **Sylow p -subgroup** of G is a subgroup of order p^n .*

Theorem 1 (Sylow's Theorem). *Let $|G| = p^n m$ with $\gcd(p, m) = 1$. Then:*

- (i) *Sylow p -subgroups exist.*
- (ii) *Any two Sylow p -subgroups are conjugate in G .*
- (iii) *The number of Sylow p -subgroups, denoted n_p , satisfies:*

$$n_p \equiv 1 \pmod{p} \quad \text{and} \quad n_p \mid m.$$

Structure of D_{12}

Definition 2 (Dihedral Group D_{12}). *The dihedral group D_{12} is the group of symmetries of a regular hexagon, consisting of 6 rotations and 6 reflections. It has order $|D_{12}| = 12$, with presentation:*

$$D_{12} = \langle r, s \mid r^6 = s^2 = e, srs^{-1} = r^{-1} \rangle.$$

The 12 elements of D_{12} are:

$$D_{12} = \{ e, r, r^2, r^3, r^4, r^5, s, sr, sr^2, sr^3, sr^4, sr^5 \}.$$

Order factorisation:

$$|D_{12}| = 12 = 2^2 \cdot 3.$$

So the relevant primes are $p = 2$ (with $p^n = 4$) and $p = 3$ (with $p^n = 3$).

Sylow 2-Subgroups of D_{12}

Here $p = 2$, $p^n = 2^2 = 4$, $m = 3$. We seek all subgroups of order 4.

Constraints on n_2 :

$$n_2 \equiv 1 \pmod{2} \quad \text{and} \quad n_2 \mid 3.$$

So $n_2 \in \{1, 3\}$.

Finding the Sylow 2-subgroups. We look for subgroups of order 4 in D_{12} . The subgroup of rotations $\langle r \rangle = \{e, r, r^2, r^3, r^4, r^5\} \cong \mathbb{Z}_6$ has a unique subgroup of order 2, namely $\{e, r^3\}$. We combine this with reflections to form groups of order 4.

The three Sylow 2-subgroups of D_{12} are:

$$P_1 = \{e, r^3, s, sr^3\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2,$$

$$P_2 = \{e, r^3, sr, sr^4\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2,$$

$$P_3 = \{e, r^3, sr^2, sr^5\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Proposition 1. P_1, P_2, P_3 are subgroups of D_{12} , each of order 4.

Proof. We verify $P_1 = \{e, r^3, s, sr^3\}$. Using the relations $r^6 = e, s^2 = e, srs^{-1} = r^{-1}$:

- $(r^3)^2 = r^6 = e$. ✓
- $s^2 = e$. ✓
- $(sr^3)^2 = sr^3 \cdot sr^3 = s(r^3s)r^3 = s \cdot sr^{-3} \cdot r^3 = s^2 \cdot e = e$. ✓
- $r^3 \cdot s = sr^{-3} = sr^3$ (since $r^{-3} = r^3$ in $\langle r \rangle \cong \mathbb{Z}_6$). So $r^3 \cdot s = sr^3 \in P_1$. ✓

Closure holds for all products; all elements are self-inverse. Hence $P_1 \leq D_{12}$. The verifications for P_2 and P_3 are analogous, replacing s with sr and sr^2 respectively. $\square$

Remark 1. Since $n_2 = 3 \neq 1$, **no Sylow 2-subgroup is normal** in D_{12} . The three subgroups P_1, P_2, P_3 are mutually conjugate.

Sylow 3-Subgroups of D_{12}

Here $p = 3, p^n = 3^1 = 3, m = 4$. We seek all subgroups of order 3.

Constraints on n_3 :

$$n_3 \equiv 1 \pmod{3} \quad \text{and} \quad n_3 \mid 4.$$

So $n_3 \in \{1, 4\}$. But $n_3 \mid 4$ and $n_3 \equiv 1 \pmod{3}$, giving $n_3 = 1$ (since $4 \not\equiv 1 \pmod{3}$).

Hence there is exactly **one** Sylow 3-subgroup, which must be normal in D_{12} .

Finding the Sylow 3-subgroup. Consider the rotation subgroup $\langle r \rangle \cong \mathbb{Z}_6$. It contains the unique subgroup of order 3:

$$Q = \langle r^2 \rangle = \{e, r^2, r^4\} \cong \mathbb{Z}_3.$$

Proposition 2. $Q = \{e, r^2, r^4\}$ is a normal subgroup of D_{12} .

Proof. Q is a subgroup since $r^2 \cdot r^2 = r^4, r^2 \cdot r^4 = r^6 = e$, and $r^4 \cdot r^4 = r^8 = r^2$; it is clearly closed and every element is invertible within Q . Since $n_3 = 1$, Q is the unique Sylow 3-subgroup and hence is normal: $gQg^{-1} = Q$ for all $g \in D_{12}$.

We verify normality directly for the generator s :

$$sr^2s^{-1} = r^{-2} = r^4 \in Q, \quad sr^4s^{-1} = r^{-4} = r^2 \in Q.$$

Since Q is also closed under conjugation by r (as r commutes with powers of r), we have $gQg^{-1} = Q$ for all $g \in D_{12}$. Hence $Q \trianglelefteq D_{12}$. $\square$

Summary

Prime p	$ P $	n_p	Sylow p -Subgroup(s)	Normal?
			$P_1 = \{e, r^3, s, sr^3\}$	
2	4	3	$P_2 = \{e, r^3, sr, sr^4\}$	No
			$P_3 = \{e, r^3, sr^2, sr^5\}$	
3	3	1	$Q = \{e, r^2, r^4\}$	Yes

To summarise:

- D_{12} has **three** Sylow 2-subgroups of order 4, each isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. They are not normal.
- D_{12} has a **unique** Sylow 3-subgroup of order 3, namely $Q = \langle r^2 \rangle \cong \mathbb{Z}_3$, which is normal in D_{12} .